

MetalSynth-Pelvis: Conditioned synthesis of osteosynthesis implants and local metal artifacts in pelvic CT scan volumes using Multi-Window Latent Diffusion

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1. Problem Statement

Segmentation networks for pelvic computed tomography (CT) learn from annotated volumes, and few annotated volumes contain osteosynthesis implants. In CTPelvic1K (Liu et al., 2021), the metal-bearing subset (CLINIC-metal) carries few bone annotations. Metal causes beam hardening, streaking and photon starvation, and these artifacts extend beyond the implant into the bone around it. Synthetic implants could supply such cases. This thesis builds and evaluates the synthesis method. It does not measure the effect on segmentation.

Three lines of work address parts of the problem. Diffusion models for lesion synthesis, such as those of Chen et al. (2024), Zhang et al. (2025) and Ramzan et al. (2026), condition on a mask and contain no mechanism or training signal for intensity changes outside the object, whereas metal changes intensity outside itself. Physics-based simulation with CatSim (De Man et al., 2007) within XCIST (Wu et al., 2022), as used in the hybrid protocol of Peters et al. (2025), places metal at random positions for training data and by expert choice for the benchmark. It does not place implants under anatomical constraints. Surgical planning pipelines (Liu et al., 2025) compute one trajectory, while clinical placements follow a distribution: on a four-grade cortical-breach scale (Smith et al., 2006), Zwingmann et al. (2009) report grade 0 in 69% of navigated and 40% of conventional iliosacral screws.

Gap. No published method combines (C1) rigid parametric implant geometry, (C2) a placement sampler that matches measured malposition distributions, and (C3) multi-window HU encoding, taken from metal artifact reduction (Wang et al., 2025), applied to synthesis, with a generation band B_δ (about 12 mm) beyond the implant mask so that streaks and beam hardening appear outside the metal.

2. Research Question and Hypothesis

Question. Compared with copy-paste insertion and with the hybrid physics protocol of Peters et al. (2025), does a placement sampler constrained by bone density, cortical clearance and corridor viability, combined with a multi-window latent diffusion model conditioned on implant geometry and B_δ , produce pelvic screws and local artifacts with higher surgical and physical coherence?

Hypothesis. Sampled placements lie closer (Wasserstein-1) to the two S1 breach-grade distributions of Zwingmann et al. (2009) than unconstrained placements. Synthesized appearance differs less from real artifacts than copy-paste insertion, and its streak amplitude matches that of the physics protocol.

3. Objectives

- Representation gate (Go/No-Go).** Round trip from HU to three windows, through the Stable Diffusion 1.5 autoencoder (Rombach et al., 2022), and back to HU, with bone mean absolute error (MAE) below 25 HU on 34 held-out patients. Two autoencoder variants (pretrained, and decoder adapted with encoder frozen) and three encodings are tested. The rule was fixed before any run. The 25 HU level follows the error of current methods on the same scale (RMSE of 12.3 to 20.2 HU in Karageorgos et al. (2024) and 12.74 HU in Yun et al. (2026)). If no combination passes, Objective 3 is not pursued and the No-Go is the result.
- Placement sampler.** 3D pose of a parametric screw (6.3 to 8 mm) in the CT reference frame of Kaiser et al. (2014), constrained by bone density read from CT numbers (Arand et al., 2019), 5 mm cortical clearance and corridor viability (McLaren et al., 2021), measured on TotalSegmentator masks (Wasserthal et al., 2023). Target: the two ordinal distributions of Zwingmann et al. (2009) at S1. S2 is reported without a clinical target.
- Synthesizer.** 2.5D ControlNet (Zhang et al., 2023) on Stable Diffusion 1.5, with frozen encoder and the generation band B_δ .
- Evaluation.** Surgical Admissibility of Placement (SAP), the only metric this work introduces, compared by Wasserstein-1 distance. Appearance uses bone integrity, metal integrity and streak amplitude under their published names (Peters et al., 2025). RMSE and SSIM outside B_δ check that no other region changes.

4. Expected Results

Goal	Metric	Expected outcome
Representation (Obj. 1)	Bone MAE in HU after the round trip, mean per patient on 34 test patients	Below 25 HU for at least one pre-specified combination, or a reported No-Go.
Placement (Obj. 2 and 4)	SAP: four-grade cortical breach (Smith et al., 2006), density-zone fraction, corridor viability (McLaren et al., 2021)	Lower Wasserstein-1 distance to the two S1 distributions of Zwingmann et al. (2009) than unconstrained placement. S2 reported without a clinical target.
Appearance (Obj. 3 and 4)	Bone integrity, metal integrity, streak amplitude (Peters et al., 2025)	Lower discrepancy than copy-paste insertion. Streak amplitude comparable to the physics protocol.
Preservation	RMSE and SSIM against the original CT, outside B_δ	No change outside B_δ .

5. Data and Metal Extraction

Data. 178 local CTPelvic1K volumes (103 in dataset6 and 75 in dataset7) from 168 patients, split by patient (126 train, 8 validation, 34 test) so that no patient enters both training and evaluation. Challenge data (Haneda et al., 2025) are not used for training.

Handling false positives in metal extraction. A threshold on HU returns voxels that are not implant: cortical bone, objects outside the body, and bright streaks next to the metal. Thresholded masks over-cover the metal (Xie et al., 2024). Four measures address this. (i) *Screening ranks, review decides.* A 2500 HU threshold selects volumes for review: it flagged 113 of 178 and missed none of the volumes with metal or foreign objects in a 3D review. At 1500 HU it flagged 177, since cortical bone exceeds that value. Each flagged volume is labelled by 3D and multiplanar review, which separates pelvic osteosynthesis, other implants and objects outside the body. A flag without a confirmed object does not enter the metal cohort. (ii) *No extracted geometry enters synthesis.* Of 57 elongated objects extracted by HU threshold, 9 were fragmented and 49 measured below 6.0 mm in diameter, under the published screw calibre. Implants are therefore parametric cylinders, and extraction errors cannot reach the generator. (iii) *Two rules bound each real metal mask.* Where a real mask is needed (space occupied by an existing screw), a local half-maximum threshold per object and a fixed 2500 HU threshold are both applied, and corridor viability fell from 53.8% to 40.4% of 52 patients under both. Metal integrity uses the threshold per region of interest defined by Peters et al. (2025). (iv) *Artifacts in anatomical masks.*

Components below 0.1% of their structure are removed and each case passes quality control. A landmark region counts as contaminated when its dark-streak or metal fraction exceeds the maximum found in 69 metal-free patients. Of 65 patients with orthopaedic metal, the reference frame was computable in 48 and free of contamination in 29. Contaminated cases are reported apart and not pooled.

6. Computational Resources and Budget

Resources. Khipu, the UTEC cluster: two nodes with one NVIDIA RTX A6000 (48 GB) each, one A100 node with MIG partitions, and a CPU node for steps without GPU. The thesis account allows 3 running jobs, 32 CPU cores, 98 GB of RAM and 24 h per job. Scripts save checkpoints and resume across jobs.

Measured costs. Anatomical segmentation of the local cohort: 6 h 27 min. Objective 1 decoder adaptation: 30,000 steps at 1.32 s per step and 24.6 GB of GPU memory. One encoding finished in 10 h 36 min, so the three encodings take about 33 GPU-hours. Evaluation takes about 65 s per volume, model and encoding.

Pending estimate. ControlNet training time is not yet measured, and no figure is given for it here. It will be fixed by the procedure used for Objective 1: a timed pilot of 200 steps on an RTX A6000, and the number of planned steps multiplied by the seconds per step, before the full run. If the Objective 1 gate fails, ControlNet is not trained. The physics arm runs on a reduced patient subset. Wu et al. (2022) report a run time of “a few minutes on a PC with 12 CPU-cores”, so the arm is planned on the CPU node, and its time per volume will be measured in its first run.

Scope limits. Downstream segmentation (Dice, HD95) and ablations per constraint are out of scope, which removes their training cost. Joint prostheses are excluded: in training sinograms generated by XCIST, the metal trace “may look distorted for large metal objects with diameters larger than 3.0 cm” (Haneda et al., 2025). This work models screws.

7. References

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